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Deep Generative Models

Lecture 13

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Recap of Previous Lecture

Pair conditioning uses $\mathbf{z} = (\mathbf{x}_0, \mathbf{x}_1)$ with a coupling $p(\mathbf{x}_0, \mathbf{x}_1)$ whose marginals are p_0 and p_1.

Choices: independent pairs ($p = p_0 p_1$), paired data, or minibatch OT matching.

Desired conditional endpoints:

$$p_0(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1) = \delta(\mathbf{x} - \mathbf{x}_0),$$
$$p_1(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1) = \delta(\mathbf{x} - \mathbf{x}_1).$$

Gaussian Conditional Probability Path

$$p_t(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1) = \mathcal{N}\left(\boldsymbol{\mu}_t(\mathbf{x}_0, \mathbf{x}_1), \boldsymbol{\sigma}_t^2(\mathbf{x}_0, \mathbf{x}_1)\right),$$
$$\mathbf{x}_t = \boldsymbol{\mu}_t(\mathbf{x}_0, \mathbf{x}_1) + \boldsymbol{\sigma}_t(\mathbf{x}_0, \mathbf{x}_1) \odot \boldsymbol{\epsilon}.$$

Here $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. Positive Gaussian scales give smooth densities; the desired Dirac endpoints require the endpoint scales to vanish.

Tong A. et al. Improving and Generalizing Flow-Based Generative Models with Minibatch Optimal Transport, 2023.
A Visual Dive into Conditional Flow Matching.

Recap of Previous Lecture

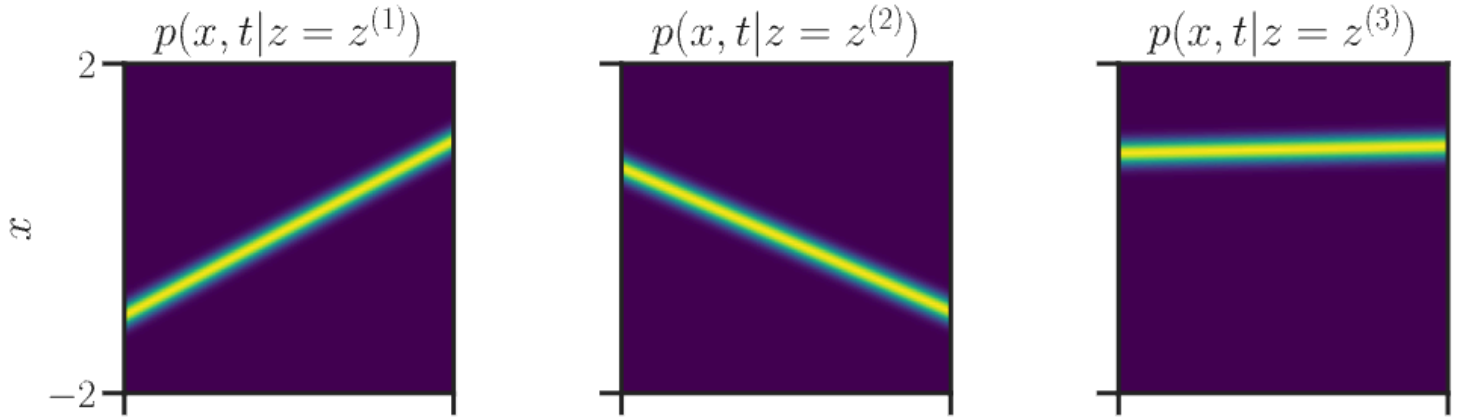
Straight conditional paths with constant Gaussian smoothing $\sigma > 0$:

$$\boldsymbol{\mu}_t = t\mathbf{x}_1 + (1-t)\mathbf{x}_0, \qquad p_t^\sigma(\mathbf{x}|\mathbf{x}_0, \mathbf{x}_1) = \mathcal{N}(\boldsymbol{\mu}_t, \sigma^2\mathbf{I}).$$

$$\mathbf{x}_t = \boldsymbol{\mu}_t + \sigma\boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \qquad \frac{d\mathbf{x}_t}{dt} = \mathbf{x}_1 - \mathbf{x}_0.$$

Hold $\boldsymbol{\epsilon}$ fixed along the path. For $i \in \{0, 1\}$, $\mathcal{N}(\mathbf{x}_i, \sigma^2\mathbf{I})$ converges weakly to $\delta(\mathbf{x} - \mathbf{x}_i)$ as $\sigma \to 0$.

The deterministic training rule uses $\sigma = 0$: $\mathbf{x}_t = \boldsymbol{\mu}_t$, with the same target $\mathbf{x}_1 - \mathbf{x}_0$.



Tong A. et al. Improving and Generalizing Flow-Based Generative Models with Minibatch Optimal Transport, 2023.
Image: A Visual Dive into Conditional Flow Matching.

Recap of Previous Lecture

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}(0))}\mathbb{E}_{t\sim U[0,1]}\mathbb{E}_{q(\mathbf{x}(t)|\mathbf{x}(0))}\left\|\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}(t),t)-\nabla_{\mathbf{x}(t)}\log q(\mathbf{x}(t)|\mathbf{x}(0))\right\|_2^2$$

$$p_t(\mathbf{x}|\mathbf{x}_1)=q_{1-t}(\mathbf{x}|\mathbf{x}_0=\mathbf{x}_1)$$

Variance Exploding SDE Probability Path

$$p_t(\mathbf{x}|\mathbf{x}_1)=\mathcal{N}\left(\mathbf{x}_1,\sigma_{1-t}^2\mathbf{I}\right)$$

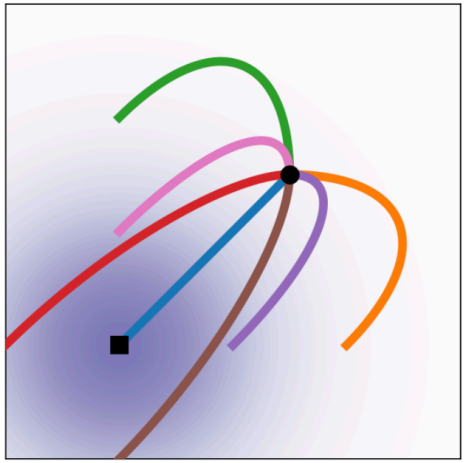
$$\Rightarrow\quad \mathbf{v}(\mathbf{x}_t,\mathbf{x}_1,t)=-\frac{\sigma'_{1-t}}{\sigma_{1-t}}(\mathbf{x}_t-\mathbf{x}_1)$$

Recap of Previous Lecture

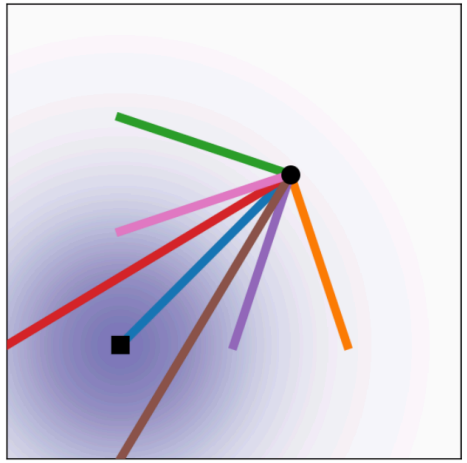
Variance Preserving SDE Probability Path

$$p_t(\mathbf{x}|\mathbf{x}_1) = \mathcal{N}\left(\alpha_{1-t}\mathbf{x}_1, (1-\alpha_{1-t}^2)\mathbf{I}\right)$$

$$\Rightarrow \quad \mathbf{v}(\mathbf{x}_t, \mathbf{x}_1, t) = \frac{\alpha'_{1-t}}{1-\alpha_{1-t}^2} \cdot (\alpha_{1-t}\mathbf{x}_t - \mathbf{x}_1)$$



Diffusion



OT

Recap of Previous Lecture

Continuous state space

- **Discrete time $t \in \{0, 1, \ldots, T\} \Rightarrow$ DDPM / NCSN**.
- **Continuous time $t \in [0, 1] \Rightarrow$ Score-based SDE models**.

Discrete state space

- **Discrete time $t \in \{0, 1, \ldots, T\}$.**
- **Continuous time $t \in [0, 1]$.**

Potential advantages over left-to-right AR

- Parallel generation
- Bidirectional context
- Flexible generation order
- Infilling and controllability
- Iterative refinement (with remasking or corrective updates)

Nie S. et al. Large language diffusion models, 2025. Wu C. et al. Fast-dLLM, 2025.
Schiff Y. et al. Simple guidance mechanisms for discrete diffusion models, 2025. Wang G. et al. Remasking discrete diffusion models with inference-time scaling, 2025.

Recap of Previous Lecture

Discrete Diffusion Markov Chain

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathrm{Cat}(\mathbf{Q}_t\mathbf{x}_{t-1}),$$

Each $\mathbf{x}_t \in \{0,1\}^K$ is a **one-hot vector** encoding the categorical state (it is just one token).

Transition Matrix

$$[\mathbf{Q}_t]_{ij} = q(x_t = i|x_{t-1} = j), \qquad \sum_{i=1}^{K}[\mathbf{Q}_t]_{ij} = 1.$$

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathrm{Cat}(\mathbf{Q}_{1:t}\mathbf{x}_0), \qquad \mathbf{Q}_{1:t} = \mathbf{Q}_t\mathbf{Q}_{t-1}\cdots\mathbf{Q}_1.$$

- The choice of $\mathbf{Q}_t$ determines how information is erased and what the stationary distribution becomes.
- $\mathbf{Q}_t$ and $\mathbf{Q}_{1:t}$ should be easy to compute for each t.

Recap of Previous Lecture

Uniform vs. Absorbing Transition Matrix

Aspect	Uniform Diffusion	Absorbing Diffusion
$\mathbf{Q}_t$	$(1-\beta_t)\mathbf{I} + \beta_t \mathbf{U}$	$(1-\beta_t)\mathbf{I} + \beta_t \mathbf{e}_m \mathbf{1}^\top$
$\mathbf{Q}_{1:t}$	$\bar{\alpha}_t \mathbf{I} + (1-\bar{\alpha}_t)\mathbf{U}$	$\bar{\alpha}_t \mathbf{I} + (1-\bar{\alpha}_t)\mathbf{e}_m \mathbf{1}^\top$
$\mathbf{Q}_{1:\infty}$, $\bar{\alpha}_t \to 0$	$\mathbf{U}$	$\mathbf{e}_m \mathbf{1}^\top$
Interpretation	Random replacement	Gradual masking of tokens
Application	Image diffusion	Text diffusion $\approx$ Masked LM

For absorbing diffusion, the limiting token distribution is $\mathrm{Cat}(\mathbf{e}_m)$.

Observation

Both schemes gradually destroy information, but differ in their stationary limit. Absorbing diffusion bridges diffusion and masked-language-model objectives.

Austin J. et al. Structured denoising diffusion models in discrete state-spaces, 2021.

Recap of Previous Lecture

Discrete conditioned reverse distribution

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \text{Cat}\left(\frac{\mathbf{Q}_t^\top \mathbf{x}_t \odot \mathbf{Q}_{1:t-1}\mathbf{x}_0}{\mathbf{x}_t^\top \mathbf{Q}_{1:t}\mathbf{x}_0}\right).$$

ELBO term

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} D_{\text{KL}}\big(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \,\|\, p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)\big).$$

- The reverse process $p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \text{Cat}\big(\boldsymbol{\pi}_{\boldsymbol{\theta}}(\mathbf{x}_t, t)\big)$ is a learned categorical distribution.
- Minimizing $\mathcal{L}_t$ w.r.t. $\boldsymbol{\theta}$ is equivalent to minimizing the cross-entropy

$$\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \text{H}\Big(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0),\, p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)\Big).$$

Austin J. et al. Structured denoising diffusion models in discrete state-spaces, 2021.

Recap of Previous Lecture

Independent Token-wise Forward Process

$$q(\mathbf{X}_t|\mathbf{X}_{t-1}) = \prod_{i=1}^{m} q(\mathbf{x}_t^i|\mathbf{x}_{t-1}^i) = \mathrm{Cat}(\mathbf{Q}_t\mathbf{X}_{t-1})$$

$$p_{\boldsymbol{\theta}}(\mathbf{X}_{t-1}|\mathbf{X}_t) = \prod_{i=1}^{m} p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}^i|\mathbf{X}_t).$$

- The forward process factorizes over positions.
- Each reverse factor predicts a token using the whole corrupted sequence $\mathbf{X}_t$.

Simplified denoising objective

$$\mathcal{L} = \sum_{t=1}^{T}\sum_{i=1}^{m} \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)}\Big[-\log p_{\boldsymbol{\theta}}(\mathbf{x}_0^i|\mathbf{X}_t)\Big].$$

Unweighted clean-token prediction (time conditioning is implicit), not the general discrete ELBO. The absorbing case uses a time-weighted loss over masked positions with a suitable reverse model.

Austin et al. Structured denoising diffusion models, 2021. Sahoo et al. Simple and effective masked diffusion language models, 2024.

Outline

01 Discrete Diffusion

Absorbing Diffusion
Continuous Time Formulation

02 Course Overview

Outline

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Absorbing Diffusion
Continuous Time Formulation

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Outline

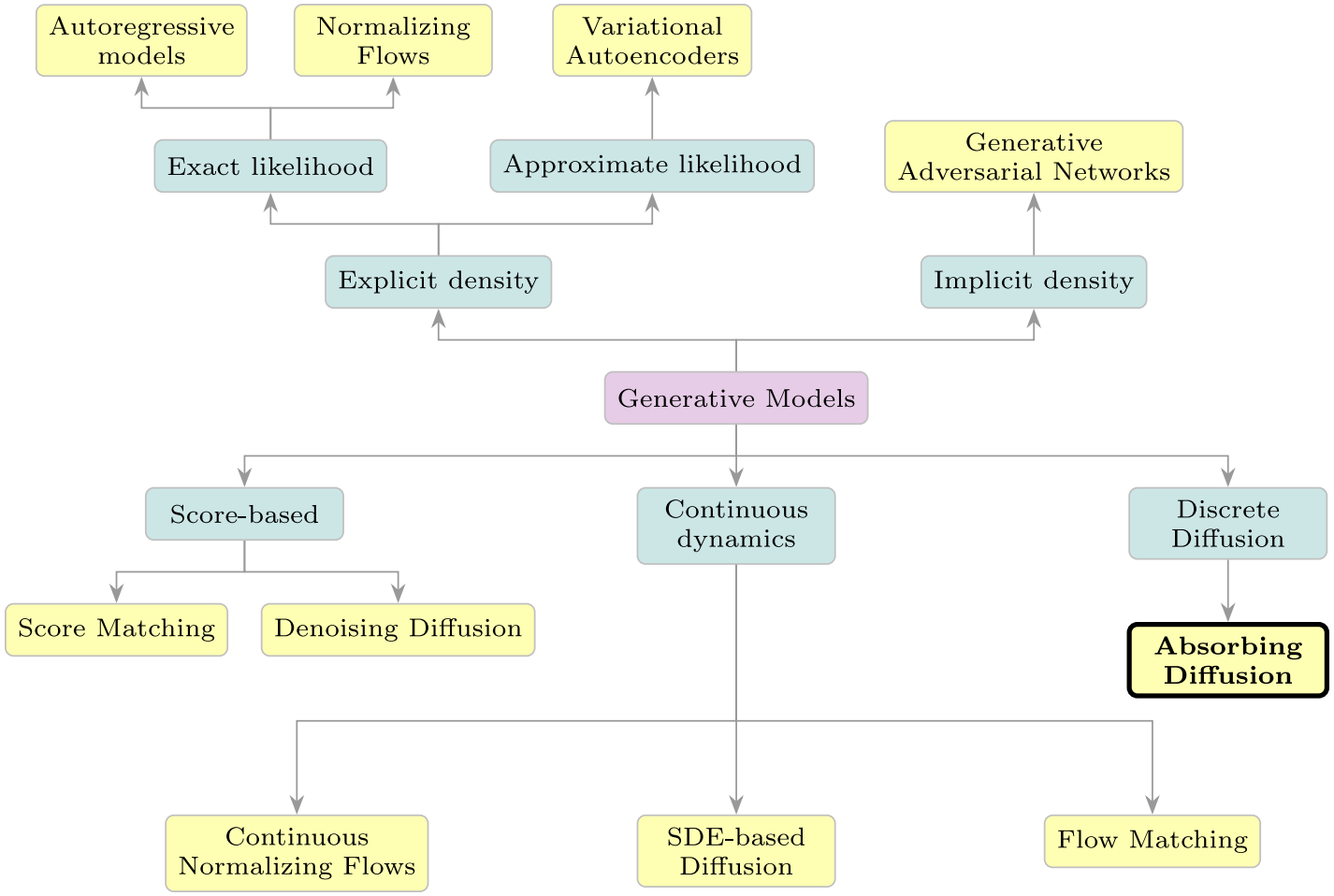
01 **Discrete Diffusion**

Absorbing Diffusion

Continuous Time Formulation

02 Course Overview

Generative Models Taxonomy



Absorbing Diffusion: Forward Process

Let's restrict to the case of absorbing transition matrix.

$$\mathbf{Q}_t = (1-\beta_t)\mathbf{I} + \beta_t \mathbf{e}_m \mathbf{1}^\top, \qquad \bar{\alpha}_t = \prod_{s=1}^{t}(1-\beta_s),$$

$$\mathbf{Q}_{1:t} = \bar{\alpha}_t \mathbf{I} + (1-\bar{\alpha}_t)\mathbf{e}_m \mathbf{1}^\top.$$

Each position is either still clean or already masked:

$$q(\mathbf{x}_t|\mathbf{x}_0) = \bar{\alpha}_t[\mathbf{x}_t = \mathbf{x}_0] + (1-\bar{\alpha}_t)[\mathbf{x}_t = \mathbf{e}_m]$$

$$\mathbf{Q}_t = \begin{pmatrix} 1-\beta_t & 0 & 0 \\ 0 & 1-\beta_t & 0 \\ \beta_t & \beta_t & 1 \end{pmatrix} \quad \Rightarrow \quad \text{the masked state is absorbing.}$$

What happens in the conditioned reverse process $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$?

Sahoo S. et al. Simple and effective masked diffusion language models, 2024

Absorbing Diffusion

Conditioned reverse distribution

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \begin{cases} [\mathbf{x}_{t-1} = \mathbf{x}_t], & \text{if } \mathbf{x}_t \neq \mathbf{e}_m, \\ \rho_t[\mathbf{x}_{t-1} = \mathbf{x}_0] + (1-\rho_t)[\mathbf{x}_{t-1} = \mathbf{e}_m], & \text{if } \mathbf{x}_t = \mathbf{e}_m, \end{cases}$$

where

$$\rho_t = \frac{\beta_t \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t}.$$

- If $\mathbf{x}_t \neq \mathbf{e}_m$, then the token must be unchanged: $\mathbf{x}_{t-1} = \mathbf{x}_t$.
- Observing an unmasked token at time t fixes the entire history: $\mathbf{x}_{t-1} = \mathbf{x}_t = \cdots = \mathbf{x}_0$.

Absorbing Diffusion

- If $\mathbf{x}_t = \mathbf{e}_m$, the previous token may be either clean or masked.
- With probability ρ_t, masking occurred exactly at step t (so $\mathbf{x}_{t-1} = \mathbf{x}_0$).
- With probability $(1 - \rho_t)$, the token was already masked earlier (so $\mathbf{x}_{t-1} = \mathbf{e}_m$).

Absorbing Diffusion

Sequence Distribution

$$q(\mathbf{X}_{t-1}|\mathbf{X}_t, \mathbf{X}_0) = \prod_{i=1}^{m} q(\mathbf{x}_{t-1}^i|\mathbf{x}_t^i, \mathbf{x}_0^i).$$

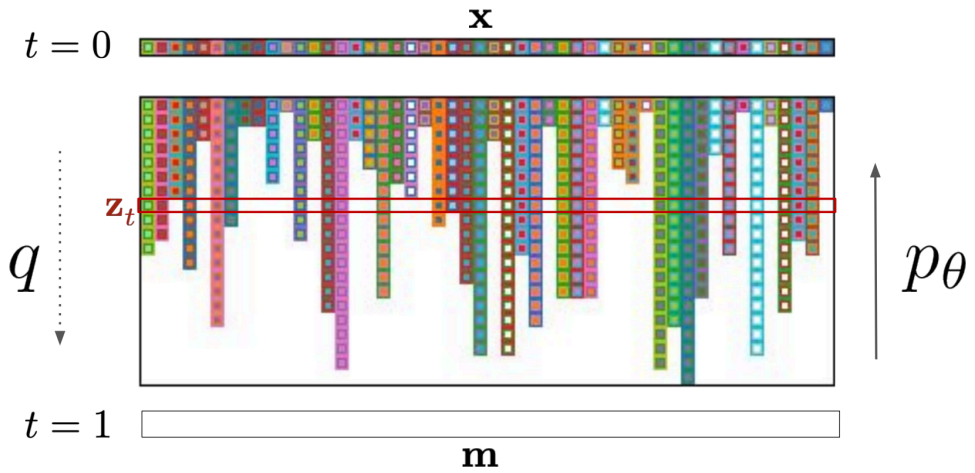
Each position i has two possible cases:

$$\mathbf{x}_t^i \neq \mathbf{e}_m \;\Rightarrow\; \mathbf{x}_{t-1}^i = \mathbf{x}_t^i, \qquad \mathbf{x}_t^i = \mathbf{e}_m \;\Rightarrow\; \mathbf{x}_{t-1}^i \in \{\mathbf{x}_0^i, \mathbf{e}_m\}.$$

Interpretation

- The forward process produces **random partial observations** of the clean sequence.
- If a token is visible at time t, the forward posterior is deterministic.
- If the reverse model copies visible tokens, their KL terms are zero. Only masked positions contribute to the training loss.

Absorbing Diffusion



$$p_{\boldsymbol{\theta}}(\mathbf{X}_{t-1}|\mathbf{X}_t) = \prod_{i=1}^{m} p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}^i|\mathbf{X}_t).$$

Copy visible tokens. At masked positions, unmask with probability ρ_t using a clean-token predictor with zero mask probability; otherwise keep the mask.

Objective: sequence-level $\mathcal{L}_t$

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{X}_t|\mathbf{X}_0)} \sum_{i=1}^{m} \rho_t[\mathbf{x}_t^i = \mathbf{e}_m] \left[-\log p_{\boldsymbol{\theta}}(\mathbf{x}_0^i|\mathbf{X}_t)\right] + \text{const}$$

Sahoo S. et al. Simple and effective masked diffusion language models, 2024

Algorithm: Discrete-Time Absorbing Diffusion

Training

1. Sample $\mathbf{X}_0 \sim p_{\text{data}}(\mathbf{X})$, $t \sim U\{1, \dots, T\}$.
2. Sample the corrupted sequence by independent masking:

$$\mathbf{x}_t^i = \begin{cases} \mathbf{e}_m, & \text{with prob. } 1 - \bar{\alpha}_t, \\ \mathbf{x}_0^i, & \text{with prob. } \bar{\alpha}_t, \end{cases} \qquad i = 1, \dots, m.$$

3. Predict token distributions in parallel:

$$p_{\boldsymbol{\theta}}(\mathbf{X}_0 | \mathbf{X}_t) = \prod_{i=1}^{m} p_{\boldsymbol{\theta}}(\mathbf{x}_0^i | \mathbf{X}_t).$$

4. Compute masked-CE loss:

$$\mathcal{L} = \rho_t \sum_{i=1}^{m} [\mathbf{x}_t^i = \mathbf{e}_m] \left[-\log p_{\boldsymbol{\theta}}(\mathbf{x}_0^i | \mathbf{X}_t) \right].$$

Algorithm: Discrete-Time Absorbing Diffusion

Sampling

1. Initialize $\mathbf{X}_T \leftarrow \mathbf{e}_m \mathbf{1}^\top$ (fully masked).
2. Iterate for $t = T, T-1, \ldots, 1$:
    1. Predict $p_{\boldsymbol{\theta}}(\mathbf{x}_0^i | \mathbf{X}_t)$ for all positions.
    2. For each masked position ($\mathbf{x}_t^i = \mathbf{e}_m$):

$$\mathbf{x}_{t-1}^i = \begin{cases} \hat{\mathbf{x}}_0^i \sim p_{\boldsymbol{\theta}}(\mathbf{x}_0^i | \mathbf{X}_t), & \text{with prob. } \rho_t, \\ \mathbf{e}_m, & \text{with prob. } 1 - \rho_t. \end{cases}$$

    3. For each unmasked position: $\mathbf{x}_{t-1}^i = \mathbf{x}_t^i$.
3. Return the final sequence $\mathbf{X}_0$ (fully unmasked).

Why Unmask in Several Steps?

Suppose the data contain only two sequences: $p_{\text{data}}(AA) = p_{\text{data}}(BB) = \frac{1}{2}$.

From `[MASK, MASK]`, even a perfect predictor assigns probability $\frac{1}{2}$ to A and B at each position.

Unmask both positions independently

Sequence	AA	AB	BA	BB
Data	$1/2$	0	0	$1/2$
One-step sampling	$1/4$	$1/4$	$1/4$	$1/4$

Half the samples are invalid: correct token marginals do not specify their joint distribution.

Reveal a token, then predict again

`[MASK, MASK]` → `[A, MASK]` → `[A, A]`. Given the first A, the perfect predictor assigns probability 1 to the second A.

Later steps condition on revealed tokens, allowing dependent choices. The absorbing sampler keeps revealed tokens fixed.

Factorized reverse process and sampling: Sahoo S. et al. Simple and Effective Masked Diffusion Language Models, 2024.

Outline

01 **Discrete Diffusion**

Absorbing Diffusion

Continuous Time Formulation

02 Course Overview

From Discrete Time to Mask Rate

In absorbing diffusion, the forward process is

$$q(\mathbf{x}_t|\mathbf{x}_0) = \bar{\alpha}_t[\mathbf{x}_t = \mathbf{x}_0] + (1 - \bar{\alpha}_t)[\mathbf{x}_t = \mathbf{e}_m].$$

- The distribution depends on t only through the scalar

$$\lambda_t = 1 - \bar{\alpha}_t \in [0, 1].$$

- We can therefore reparameterize the corruption level by

$$t \quad \Rightarrow \quad \lambda \in [0, 1].$$

- We directly define a family of corrupted distributions indexed by a continuous mask rate λ:

$$q(\mathbf{x}_\lambda|\mathbf{x}_0) = (1 - \lambda)[\mathbf{x}_\lambda = \mathbf{x}_0] + \lambda[\mathbf{x}_\lambda = \mathbf{e}_m].$$

Discrete ELBO Revisited

Recall the per-step ELBO term for absorbing diffusion:

$$\mathcal{L}_t = \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \sum_{i=1}^{m} \rho_t[\mathbf{x}_t^i = \mathbf{e}_m] \left[-\log p_{\boldsymbol{\theta}}(\mathbf{x}_0^i|\mathbf{X}_t)\right] + \text{const}.$$

Replacing the discrete index t with the continuous mask rate λ, the training objective becomes

$$\mathcal{L} = \int_0^1 w(\lambda) \mathbb{E}_{q_\lambda(\mathbf{x}_\lambda|\mathbf{x}_0)} \sum_{i=1}^{m} [\mathbf{x}_\lambda^i = \mathbf{e}_m] \left[-\log p_{\boldsymbol{\theta}}(\mathbf{x}_0^i|\mathbf{X}_\lambda)\right] d\lambda.$$

Interpretation

Training corresponds to optimizing a **continuous mixture of masked language modeling objectives** with different mask rates.

Algorithm: Masked Diffusion Language Model (MDLM)

Training

1. Sample $\mathbf{X}_0 \sim p_{\text{data}}(\mathbf{X})$, $\lambda \sim U[0, 1]$.
2. Sample the corrupted sequence by independent masking:

$$\mathbf{x}_\lambda^i = \begin{cases} \mathbf{e}_m, & \text{with prob. } \lambda, \\ \mathbf{x}_0^i, & \text{with prob. } 1 - \lambda, \end{cases} \qquad i = 1, \ldots, m.$$

3. Predict token distributions in parallel:

$$p_{\boldsymbol{\theta}}(\mathbf{X}_0 | \mathbf{X}_\lambda) = \prod_{i=1}^{m} p_{\boldsymbol{\theta}}(\mathbf{x}_0^i | \mathbf{X}_\lambda).$$

4. Compute masked-CE loss:

$$\mathcal{L} = w(\lambda) \sum_{i=1}^{m} [\mathbf{x}_\lambda^i = \mathbf{e}_m] \left[-\log p_{\boldsymbol{\theta}}(\mathbf{x}_0^i | \mathbf{X}_\lambda) \right].$$

Algorithm: Masked Diffusion Language Model (MDLM)

Sampling

1. Initialize $\mathbf{X}_{\lambda_1} \leftarrow \mathbf{e}_m \mathbf{1}^\top$ (fully masked).
2. Iterate for $\ell = 1, \dots, L-1$ over a decreasing schedule $1 = \lambda_1 > \lambda_2 > \cdots > \lambda_L = 0$:
    1. Predict $p_{\boldsymbol{\theta}}(\mathbf{x}_0^i | \mathbf{X}_{\lambda_\ell})$ for all positions.
    2. For each masked position ($\mathbf{x}_{\lambda_\ell}^i = \mathbf{e}_m$):

$$\mathbf{x}_{\lambda_{\ell+1}}^i = \begin{cases} \hat{\mathbf{x}}_0^i \sim p_{\boldsymbol{\theta}}(\mathbf{x}_0^i | \mathbf{X}_{\lambda_\ell}), & \text{with prob. } 1 - \frac{\lambda_{\ell+1}}{\lambda_\ell}, \\ \mathbf{e}_m, & \text{with prob. } \frac{\lambda_{\ell+1}}{\lambda_\ell}. \end{cases}$$

    3. For each unmasked position: $\mathbf{x}_{\lambda_{\ell+1}}^i = \mathbf{x}_{\lambda_\ell}^i$.
3. Return the final sequence $\mathbf{X}_{\lambda_L}$ (fully unmasked).

Algorithm: Masked Diffusion Language Model (MDLM)

Note: Once unmasked, a token stays unmasked. So for a currently masked token:

$$\mathbb{P}(\mathbf{x}^i_{\lambda_{\ell+1}} = \mathbf{e}_m \mid \mathbf{x}^i_{\lambda_\ell} = \mathbf{e}_m) = \frac{\lambda_{\ell+1}}{\lambda_\ell}.$$

Outline

01 Discrete Diffusion

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02 Course Overview

Course Overview: Problem Statement

Goal

Learn a generative model $p_{\boldsymbol{\theta}}(\mathbf{x})$ that matches the data distribution $p_{\text{data}}(\mathbf{x})$.

Three lenses on the same problem

- **Divergence minimization** (L1, L5):

$$\min_{\boldsymbol{\theta}} D\left(p_{\text{data}} \,\|\, p_{\boldsymbol{\theta}}\right) \qquad (\text{KL, JS, Wasserstein, } \dots)$$

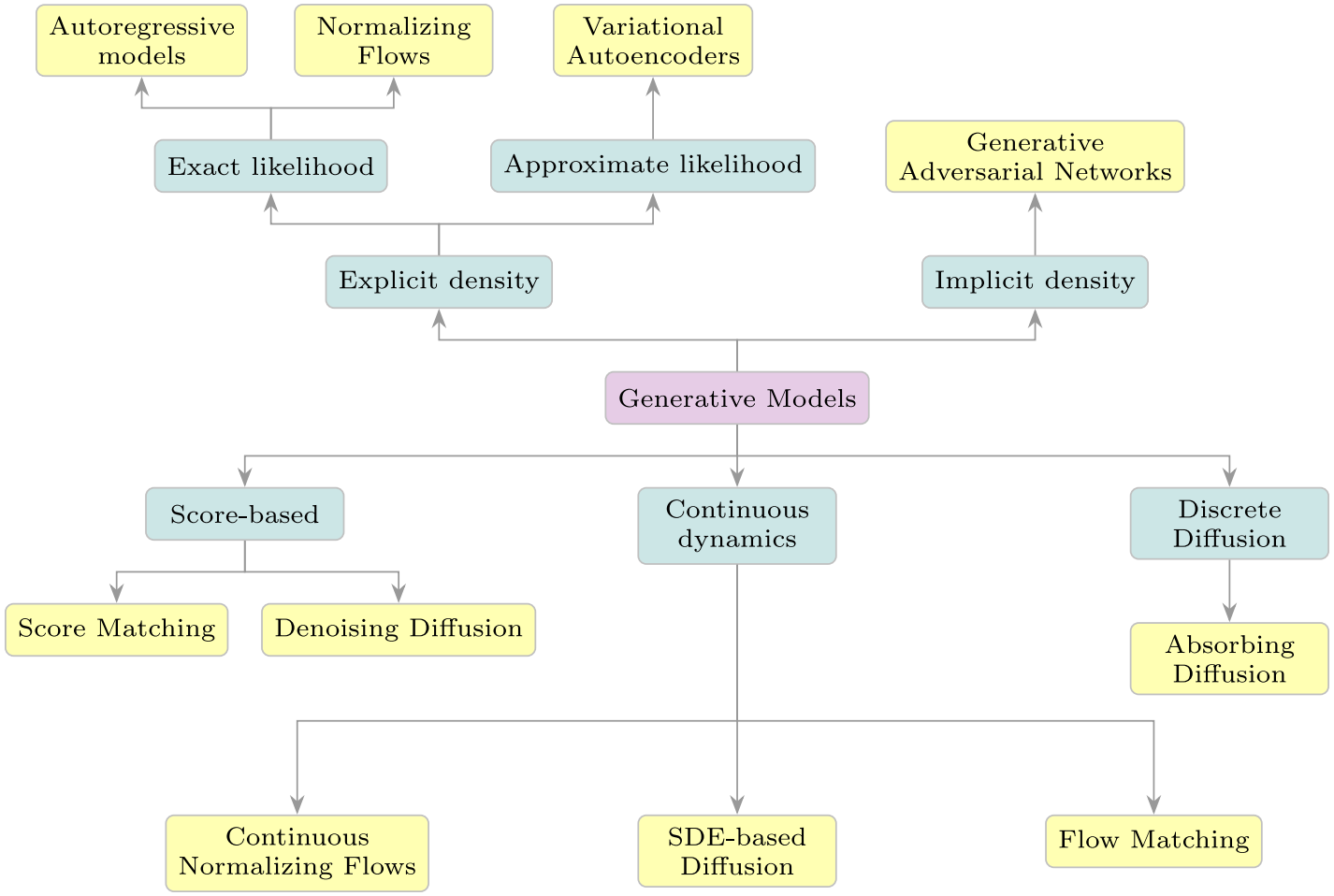
- **Likelihood-based** (L1–L4, L7–L8): maximize $\log p_{\boldsymbol{\theta}}(\mathbf{x})$

  (AR, NF, VAE, diffusion as hierarchical VAE).

- **Score-based** (L6–L10): learn $\nabla_{\mathbf{x}} \log p(\mathbf{x})$

  (SM / NCSN, DDPM, score-based SDE).

Course Roadmap



Likelihood-based: Exact & Approximate Density (L1–L4)

Autoregressive (L1)

$$p_{\boldsymbol{\theta}}(\mathbf{x}) = \prod_{i=1}^{m} p_{\boldsymbol{\theta}}(x_j | \mathbf{x}_{1:j-1}) \qquad \text{exact likelihood, sequential sampling}$$

Normalizing Flows (L2)

$$\mathbf{x} = \mathbf{f}_{\boldsymbol{\theta}}(\mathbf{z}), \qquad \log p_{\boldsymbol{\theta}}(\mathbf{x}) = \log p(\mathbf{z}) + \log \left| \det \frac{\partial \mathbf{z}}{\partial \mathbf{x}} \right|$$

VAE / VQ-VAE (L3–L4)

$$\log p_{\boldsymbol{\theta}}(\mathbf{x}) \geq \mathbb{E}_{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})} \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}) - D_{\mathrm{KL}}\big(q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}) \,\|\, p(\mathbf{z})\big)$$

- Amortized inference + reparametrization trick.
- ELBO surgery $\Rightarrow$ optimal prior; VQ-VAE with discrete latents.

Implicit & Score-based Foundations (L5–L6)

GAN / WGAN (L5)

- Implicit $p_{\boldsymbol{\theta}}$, adversarial min-max

$$\min_{\boldsymbol{\theta}} \max_{\boldsymbol{\phi}} \; \mathbb{E}_{p_{\text{data}}} \log D_{\boldsymbol{\phi}}(\mathbf{x}) + \mathbb{E}_{p(\mathbf{z})} \log\big(1 - D_{\boldsymbol{\phi}}(G_{\boldsymbol{\theta}}(\mathbf{z}))\big)$$

- Optimal discriminator $\Rightarrow$ JSD; WGAN uses Wasserstein distance via Kantorovich–Rubinstein duality.
- Evaluation tools: FID, Precision/Recall, CLIP score, human eval.

Score Matching / NCSN (L6)

- Learn $\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}) \approx \nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x})$ via denoising SM.
- Sample with (annealed) Langevin dynamics over multiple noise scales.

Diffusion Models (L7–L8)

Gaussian diffusion (L7)

- Forward $q(\mathbf{x}_t|\mathbf{x}_{t-1})$, direct sampling $q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1-\bar{\alpha}_t)\mathbf{I})$.
- Diffusion as hierarchical VAE; ELBO splits into reconstruction + prior KL + $\sum_t \mathcal{L}_t$.

DDPM (L8)

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{t,\mathbf{x}_0,\boldsymbol{\epsilon}} \left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_{\boldsymbol{\theta}}(\mathbf{x}_t, t) \right\|^2, \qquad \boldsymbol{\epsilon}_{\boldsymbol{\theta}} \propto -\sigma_t \mathbf{s}_{\boldsymbol{\theta}}$$

- Equivalent to score-based generative modeling (NCSN-style).

Guidance (L8)

$$\hat{\boldsymbol{\epsilon}}_{\boldsymbol{\theta}}(\mathbf{x}_t, t, \mathbf{y}) = (1-\gamma)\, \boldsymbol{\epsilon}_{\boldsymbol{\theta}}(\mathbf{x}_t, t, \varnothing) + \gamma\, \boldsymbol{\epsilon}_{\boldsymbol{\theta}}(\mathbf{x}_t, t, \mathbf{y})$$

Continuous Dynamics & SDE (L9–L10)

Neural ODE / continuous-time NF (L9)

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}(t), t), \qquad \frac{d}{dt}\log p_t(\mathbf{x}(t)) = -\operatorname{tr}\left(\frac{\partial \mathbf{v}_{\boldsymbol{\theta}}}{\partial \mathbf{x}}\right)$$

Forward SDE (L9–L10)

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)\, dt + g(t)\, d\mathbf{w}$$

- KFP equation; VE / VP SDEs unify NCSN and DDPM.

Reverse SDE & Probability flow ODE (L10)

$$d\mathbf{x} = \left[\mathbf{f}(\mathbf{x}, t) - g^2(t)\, \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}, t)\right] dt + g(t)\, d\bar{\mathbf{w}}$$

$$\frac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x}, t) - \tfrac{1}{2} g^2(t)\, \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}, t) \quad \text{(deterministic equivalent)}$$

Flow Matching (L10–L11)

Flow Matching objective (L10)

$$\mathcal{L}_{\text{FM}} = \mathbb{E}_{t, p_t(\mathbf{x})} \left\| \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t) - \mathbf{v}(\mathbf{x}, t) \right\|^2$$

Target $\mathbf{v}(\mathbf{x}, t)$ is the marginal vector field — intractable in general.

Conditional Flow Matching (L11)

$$\mathcal{L}_{\text{CFM}} = \mathbb{E}_{t, p(\mathbf{z}), p_t(\mathbf{x}|\mathbf{z})} \left\| \mathbf{v}_{\boldsymbol{\theta}}(\mathbf{x}, t) - \mathbf{v}(\mathbf{x}, \mathbf{z}, t) \right\|^2$$

- $\nabla_{\boldsymbol{\theta}} \mathcal{L}_{\text{FM}} = \nabla_{\boldsymbol{\theta}} \mathcal{L}_{\text{CFM}}$ — regress against *conditional* field.
- Conditioning on endpoint $\mathbf{z} = \mathbf{x}_1$ (one-sided) or pair $(\mathbf{x}_0, \mathbf{x}_1)$ (two-sided).
- Straight Gaussian paths $\Rightarrow$ closed-form $\mathbf{v}(\mathbf{x}, \mathbf{z}, t) = \mathbf{x}_1 - \mathbf{x}_0$.

Discrete Diffusion (L12–L13)

Forward discrete process (L12)

Categorical $\mathbf{x}_t$ with transition matrices $\mathbf{Q}_t$:

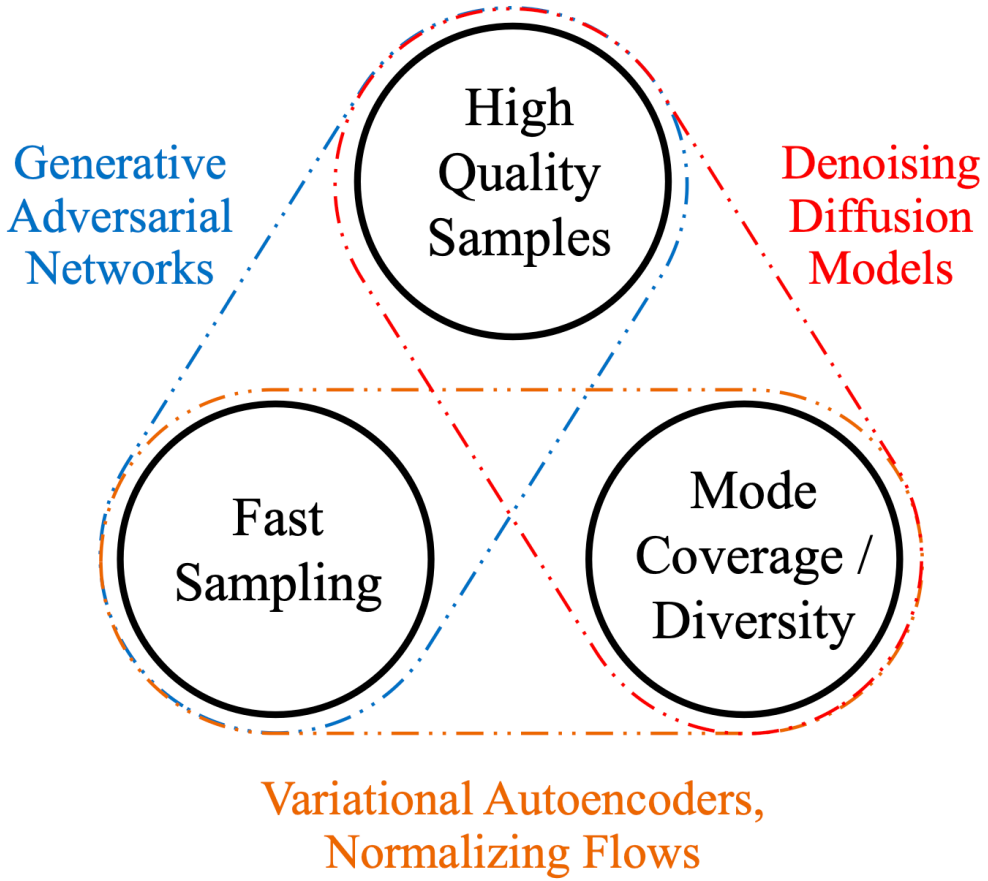
- **Uniform**: $\mathbf{Q}_t = (1-\beta_t)\mathbf{I} + \beta_t \mathbf{U}$
- **Absorbing**: $\mathbf{Q}_t = (1-\beta_t)\mathbf{I} + \beta_t \mathbf{e}_m \mathbf{1}^\top$ (mask token)

Reverse process & Discrete ELBO (L12)

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathrm{Cat}\left(\frac{\mathbf{Q}_t^\top \mathbf{x}_t \odot \mathbf{Q}_{1:t-1}\mathbf{x}_0}{\mathbf{x}_t^\top \mathbf{Q}_{1:t}\mathbf{x}_0}\right),$$

each $\mathcal{L}_t$ reduces to a **cross-entropy** loss against $p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}|\mathbf{x}_t)$.

Generative Learning Trilemma



Xiao Z., Kreis K., Vahdat A. Tackling the generative learning trilemma with denoising diffusion GANs, 2021

Generative Learning Trilemma

Rule of thumb

- **Likelihood + Coverage ⇒ AR / NF**
  exact density, *slow sampling*.
- **Likelihood + Fast Sampling ⇒ VAE**
  tractable bound, *blurry samples*.
- **Quality + Fast Sampling ⇒ GAN**
  sharp, *mode collapse, no likelihood*.
- **Quality + Coverage ⇒ Diffusion**
  stable training, *slow sampling*.
- **Quality + Faster Sampling ⇒ FM / ODE**
  fewer steps, *approx. likelihood*.
- **Discrete + Coverage ⇒ Discrete diffusion**
  parallel denoising, *iterative decoding*.

Xiao Z., Kreis K., Vahdat A. Tackling the generative learning trilemma with denoising diffusion GANs, 2021

Summary

- With a reverse model that copies visible tokens and matches the unmasking probability, the absorbing ELBO yields time-weighted losses over masked positions.

- MDLM reveals tokens in stages so later predictions can use revealed context. Independent one-step sampling can miss token dependencies, even with correct marginals.

- No generative model dominates all axes of the trilemma — each family trades off likelihood, sample quality, mode coverage and sampling speed.

- Modern frontier (flow matching, discrete diffusion, distilled diffusion) tries to push the Pareto frontier of this trilemma.